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Objective:

To understand equal and unequal IP subnetting

Question 1

Available subnet 192.168.4.0/24
Create 8 subnets

1. Write down the new subnet mask
2. Write down the number of Hosts in one subnet
3. Write down the subnet ranges

Question 2

Available subnet 148.16.0.0/16
Create 60 subnets

1. Write down the new subnet mask
2. Write down the number of Hosts in one subnet
3. Write down the subnet ranges

Question 3

Available subnet 20.6.9.0/24
Create 7 subnets

1. Write down the new subnet mask
2. Write down the number of Hosts in one subnet
3. Write down the subnet ranges

Question 4

Suppose a network with IP Address 192.16.0.0/24 is divided into 2 subnets, find number of hosts per subnet.

Also, for the first subnet, find

1. Subnet mask
2. Network address
3. First host ID
4. Last host ID
5. Broadcast Address

Question 5

For the given IP address 205.150.65.0/26 with CIDR notation find out the following

1. Subnet mask
2. No. of subnets
3. No. of hosts per subnet
4. Network ID of first subnet
5. Broadcast ID of first subnet

Question 6

You are given a block with IP address 205.150.65.0/24. You are required to create 10 subnets. Find the subnet mask.

Question 7

The network ID is 192.168.4.0/24. It's required to create three separate networks for a coffee shop named "Java Cafe". One is for the office; one is for the front desk and storage room and the other one is for public use. You need to list

1. Each network ID
2. Subnet mask
3. Host ID range
4. No of usable host IDs
5. Broadcast ID

Question 8

Subnet the IP address 196.10.20.0 into 52 usable hosts in each subnet. Find the new subnet mask.

Question 9

Subnet the IP address 216.21.5.0 into 30 usable hosts in each subnet. Find the new subnet mask.

Question 10

Available subnet 18.10.0.0/24

Required 64 usable hosts per subnet.

Find out

1. New subnet mask
2. Number of sub networks
3. Subnet range

Question 11

In a class B, network on the internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts per subnet?

Question 12

What is the broadcast address of the network 192.168.99.0 255.255.255.248?

Question 13

How many subnets and hosts per subnet can you get from the network 172.22.0.0/23?

Question 14

For IP address 192.168.1.0/24 create four subnetworks with different usable host sizes. Use concept of unequal subnetting.

1st subnet → 46 hosts

2nd subnet → 16 hosts

3rd subnet → 10 hosts

4th subnet → 2 hosts